VLSI **Design Automation** Lab

超大型積體電路(實體)**設計自動化**實驗室

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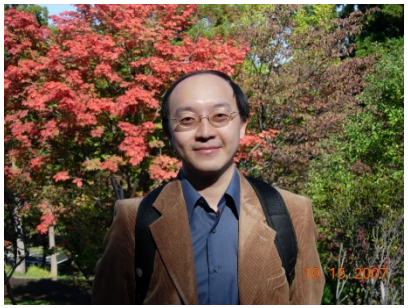
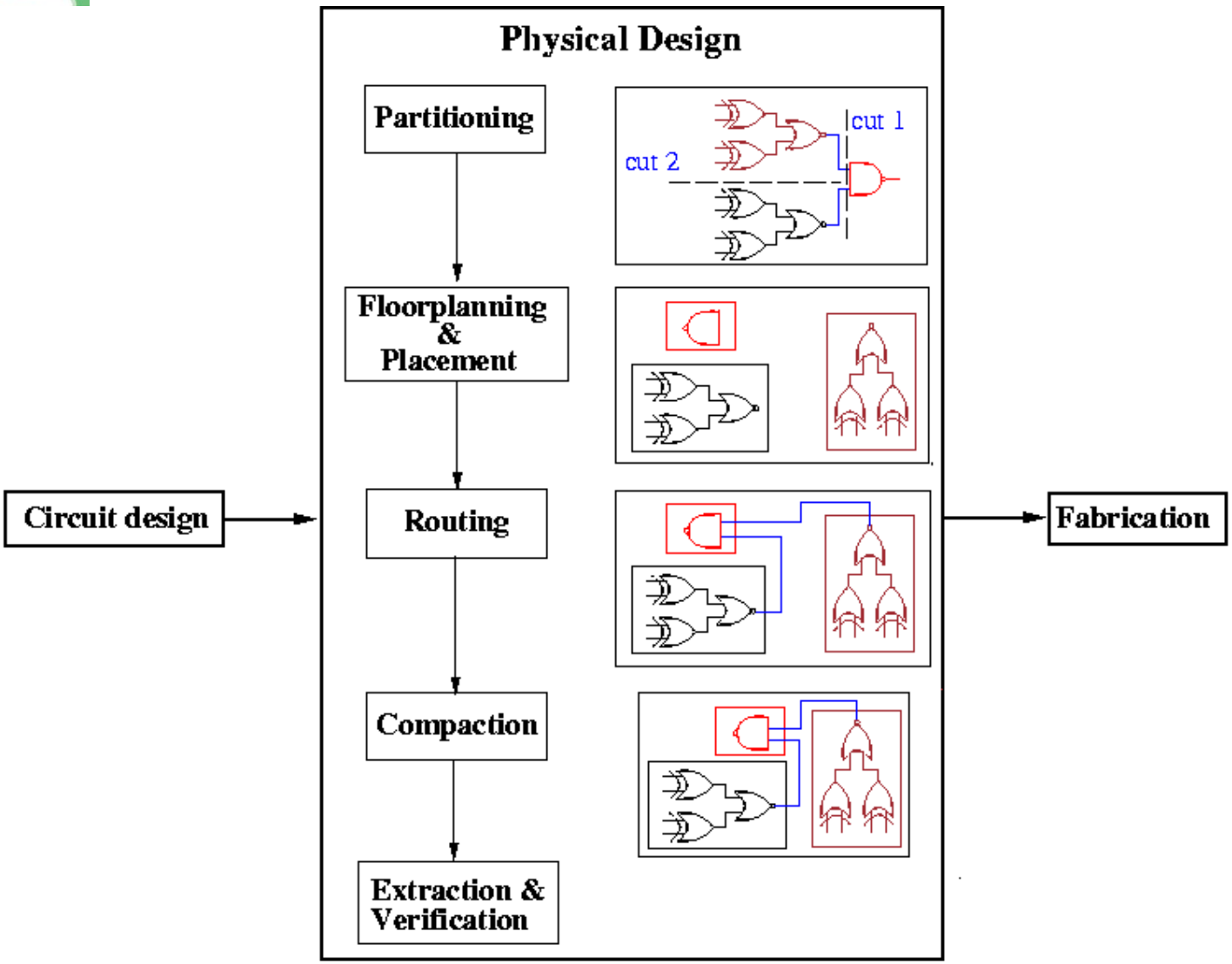
Self Introduction



- Intel internship, 1999
- Ph.D. @ UTCS, joined NCTU EE, 2003
- Visiting Research Professor
  - University of Kitakyushu, Japan, 2008-2012
  - Synopsys, Hillsboro, OR, USA, 2010
  - University of Illinois at Urbana-Champaign (UIUC), USA, 2015
- TPC members in ICCAD, ISPD, ASPDAC
- Director, Institute of Electronics, 2018-2021

Physical Design Flow



國際賽得獎榮譽

- First Prize, ACM/IEEE ICCAD CAD Contest “Static IR Drop Estimation Using Machine Learning” (with T. Liu et al.), 2023
- Fourth Place, ACM ISPD FPGA Placement Contest, 2016 and 2017
- First Prize, ACM CADathlon (with Wang and Hu), 2014
- First Prize, ACM ISPD “Detailed Routing-Driven Placement Contest” (with S. Liu et al.), 2014
- Second Prize, ACM/IEEE ICCAD CAD Contest “Synopsys Platform Development” (with Huang et al.), 2014
- Second Prize, ACM/IEEE ICCAD CAD Contest “Incremental Timing Driven Placement” (with W.-C. Wu et al.), 2014
- Second Prize, ACM CADathlon (with Chin and Wang), 2012

Welcome to the CADathlon @ ICCAD

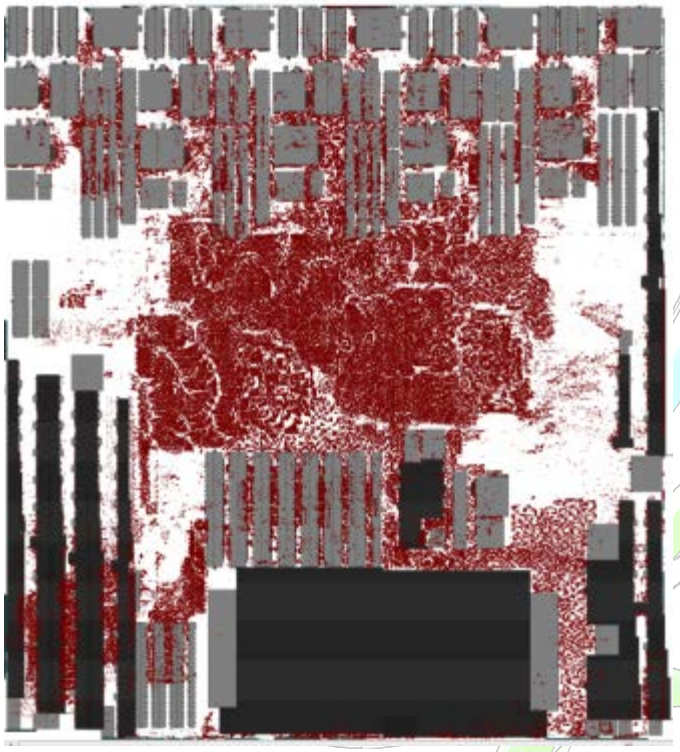
Congratulations to the 2012 Winners

First Place: Hung-Chih Ou and Po-Ya Hsu (National Taiwan University); Advisor: Yao-Wen Chang

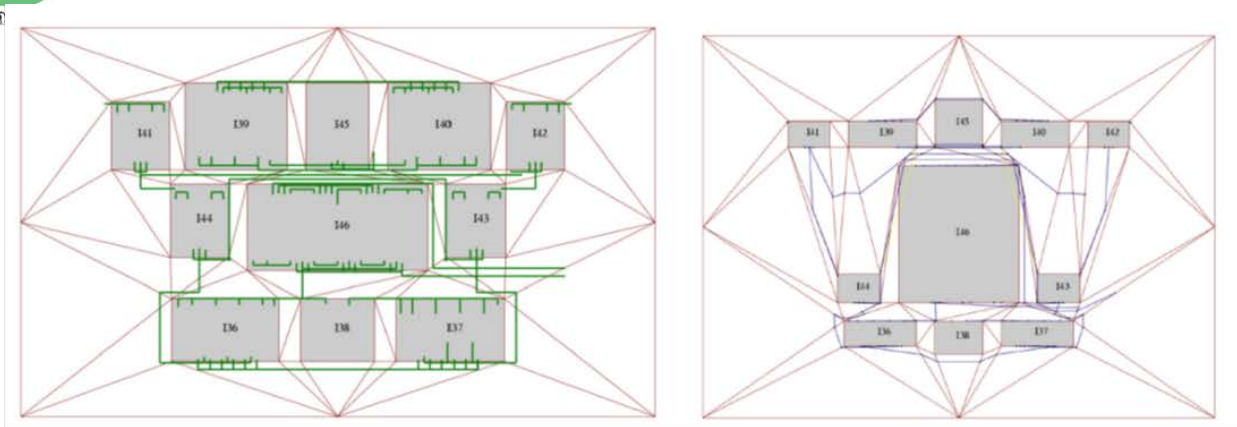
Second Place: Ching-Yu Chin and Chun-Kai Wang (National Chiao Tung University); Advisor: Hung-Ming Chen

Award Winning Placer

- ACM ISPD 2014 World Champion and 4th place FPGA placement contest
- Framework
  - SimPL based engine
  - Parallelized
- Applications
  - Thermal placement
  - FPGA heterogeneous routability-driven placement

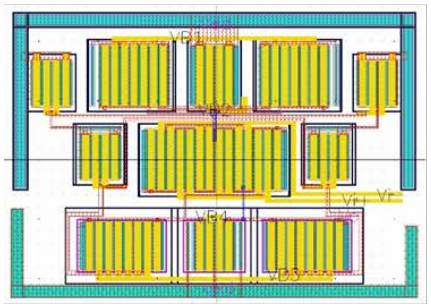


Analog Layout Migration and Generation

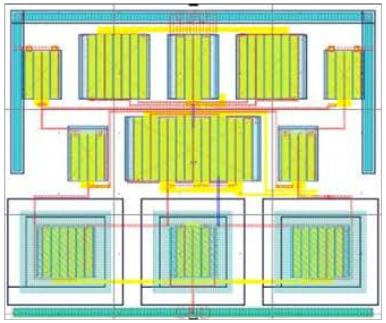


Original placement and routing with CDT extracted.(umc90)

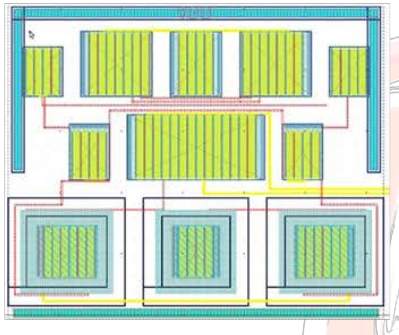
Target placement and recovered routing.(tsmc90)



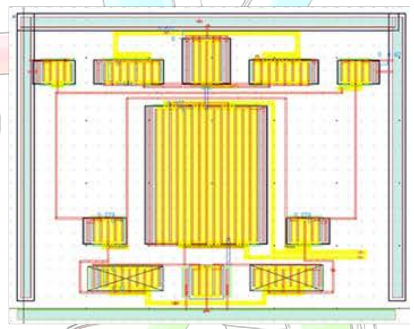
Opu90Manual



Opu65Manual



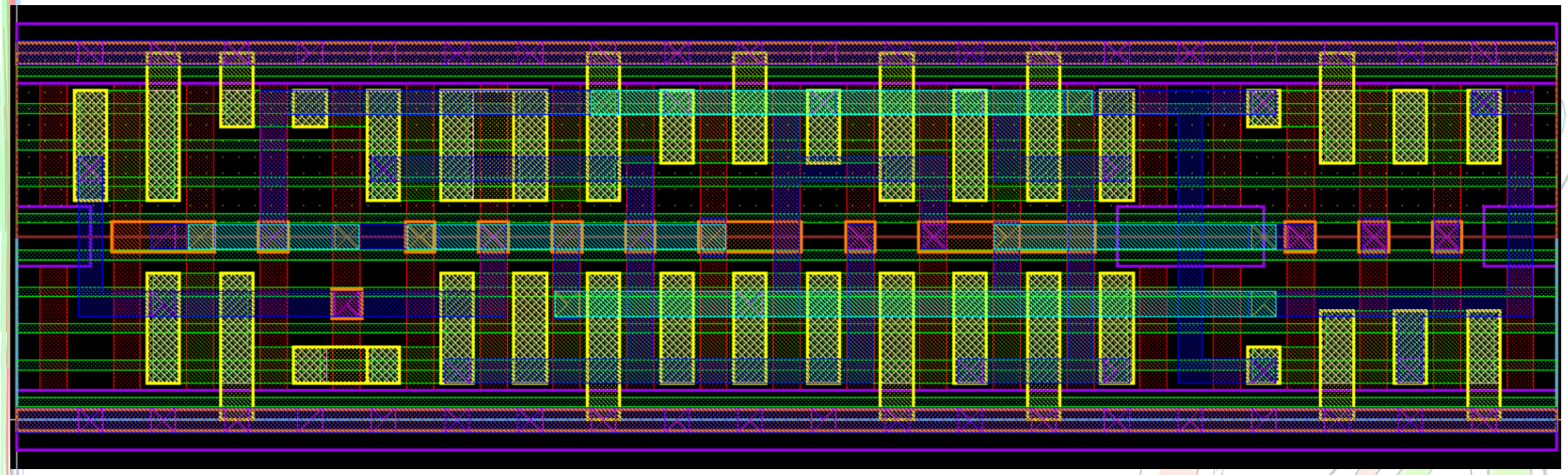
Opu65 Preserved



Opt90 Preserved

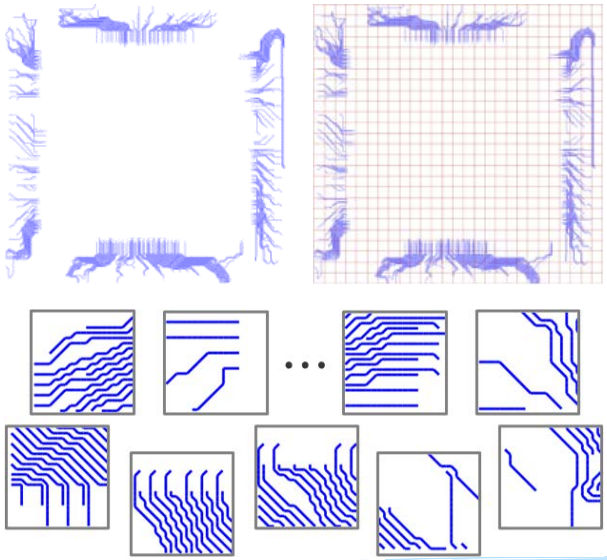
Advanced Cell Lib Generation

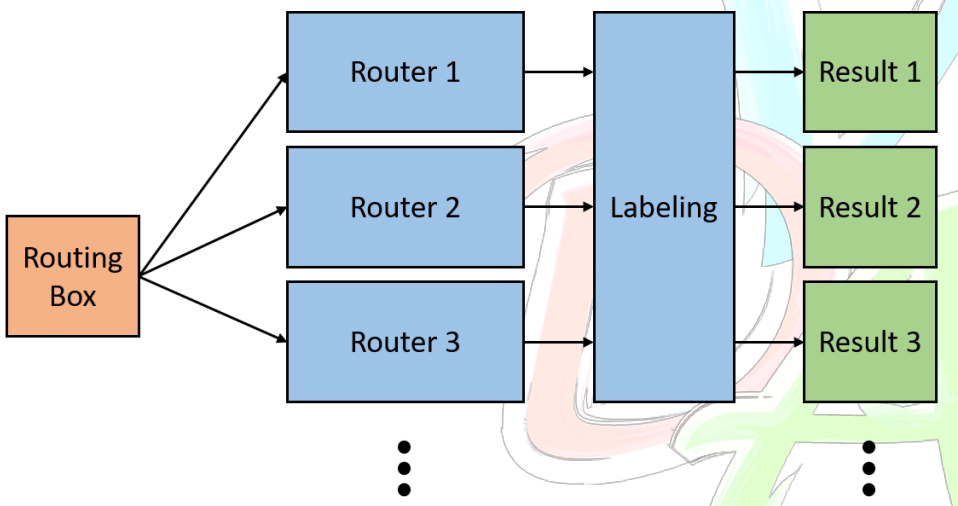
- ICGx3@ASAP7nm



AI and ML/DL in EDA

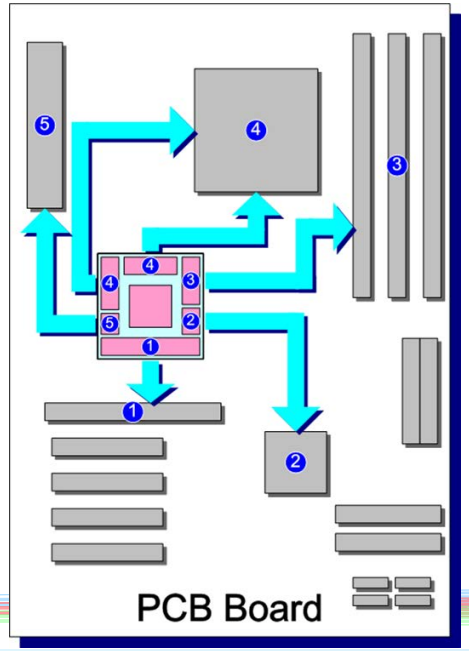
- New methodologies from AI and machine learning
- Currently applied on noise sensor placement and routing/DRC violation predition
- Next steps include analog circuit design automation and package router

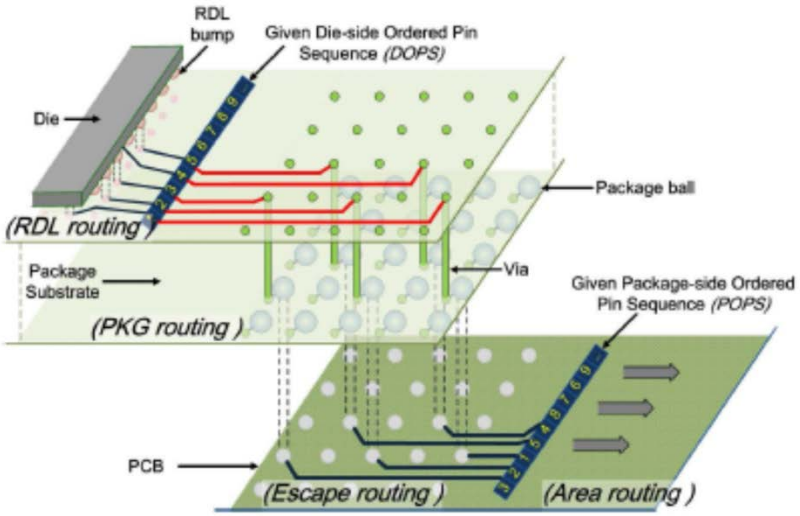




Heterogeneous Integration
Chip/Package/Board Codesign

- 2.5D/3D integration power network synthesis and cosimulation
- Board aware pinout and board level routing
- Chip/Package decap insertion and optimization





Funding Sources
委託研究經費

- Continuous supports by MOST (科技部)
- “Rapid Prototyping and Routing for Track-Based Custom Design with FinFET” and “Power Integrity Optimization for Advanced Node” by TSMC
- “3D IC Power Network Synthesis Modeling and Synthesis,” by ITRI
- “LDO Design and Automation” and “Irregular Style Package Router” by MediaTek
- “SRAM Peripheral Circuit Design Automation” by Etron
- “Partition for Multi-FPGA Systems” by Synopsys

People

- Current
  - 1 PhD students (co-advised with Prof. Liu) + 8-10 master students each year
- Alumni
  - Master: 100+
  - PhD: 10+
  - 畢業工作: 新思、益華(EDA) 、TSMC (晶圓廠, EDA) 、聯發科、聯詠、瑞昱 (IC設計) 、創意 (IC設計服務)等
- NCTU EDA group

What Techniques Do We Need in EDA Business?
攻讀EDA需要甚麼技巧呢?

- The best of both worlds
- Software background
  - Algorithms and programming language
  - Data structures play important roles
- Hardware background
  - Device modeling
  - Interconnect modeling
  - Manufacture related issues
  - Circuit design and synthesis
- Combinatorial optimization
  - Unconstrained and constrained optimization
  - Mathematical programming
- Validation and verification techniques
  - Formal methods

歡迎加入我們！

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